

Bluestock Mutual Fund Analytics Platform

Final Capstone Project Report

Bluestock Fintech | Capstone Project 2025-26

Data Scale: 40 Schemes | 46,000+ NAV Records | 32,778 Transactions

Tech Stack: Python | PostgreSQL | FastAPI | Streamlit | Plotly

1. Executive Summary

This report presents the findings from the Bluestock Mutual Fund Analytics capstone project. The platform ingests, cleans, and analyzes data for 40 real AMFI mutual fund schemes across 10 Asset Management Companies (AMCs) including SBI, HDFC, ICICI Prudential, Nippon India, Kotak, Axis, Aditya Birla, UTI, Mirae Asset, and DSP.

Key achievements:

- Built a robust ETL pipeline processing 46,000+ daily NAV records and 32,778 investor transactions.
- Implemented a star-schema database (PostgreSQL/Neon) with 2 dimension tables and 6 fact tables.
- Computed institutional-grade metrics: Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR, CVaR, CAGR, Maximum Drawdown, and a composite Fund Scorecard (0-100).
- Deployed an interactive Streamlit dashboard with 6 analytical pages.
- Built a FastAPI authentication gateway with Google OAuth and JWT session management.
- Created a premium React landing page and comprehensive PowerBI data connector.

2. Data Pipeline Architecture

The data pipeline follows a traditional ETL (Extract, Transform, Load) pattern:

EXTRACT: Raw data sourced from 10 CSV datasets covering fund master data, daily NAV history, AUM by fund house, monthly SIP inflows, category-wise net flows, industry folio counts, scheme performance metrics, investor transactions, portfolio holdings, and benchmark indices. Additionally, live NAV data is fetched from the MFAPI.in API for 6 selected schemes.

TRANSFORM: The ETL pipeline applies targeted cleaning for each dataset:

- nav_history: Multi-format date parsing, duplicate removal, NAV > 0 validation, forward-fill for holidays.
- investor_transactions: Transaction type normalization, KYC status validation, amount verification.
- scheme_performance: Anomaly flagging for returns > 200% or < -80%.

All datasets undergo generic cleaning: date parsing, numeric coercion, duplicate removal.

LOAD: Cleaned data is loaded into a PostgreSQL (Neon) cloud database using a star schema:

- Dimension tables: dim_fund (40 schemes), dim_date
- Fact tables: fact_nav, fact_transactions, fact_performance, fact_aum, fact_sip, fact_category_inflows, fact_folio, fact_holdings, fact_benchmark

3. Key Findings

3.1 Market Overview:

- Equity-oriented AUM growth saw significant increases between 2023 and 2025 (+41%).
- Monthly SIP inflows successfully breached the Rs. 31,002 Cr milestone in December 2025.
- Total industry folios crossed 2.1x growth over the analysis period.
- Mid-Cap funds generated +3.2% higher alpha compared to Large-Cap equivalents over 3 years.

3.2 Performance & Risk Insights:

- Direct plan funds consistently outperformed Regular plans due to lower expense ratios.
- Small-Cap funds showed highest annualized volatility but also highest CAGR potential.
- Large-Cap funds maintained Beta close to 1.0, confirming market-tracking behavior.
- Value-oriented funds demonstrated superior risk-adjusted returns (Sharpe > 1.0).

3.3 Investor Demographics:

- SIP mode dominates over Lumpsum for the 25-35 age group.
- T30 (Top 30 cities) account for ~70% of total transaction volume.
- KYC compliance rate is consistently above 95% across all tiers.
- Maharashtra, Karnataka, and Delhi NCR lead in both investment and redemption volumes.

4. Visual Analytics

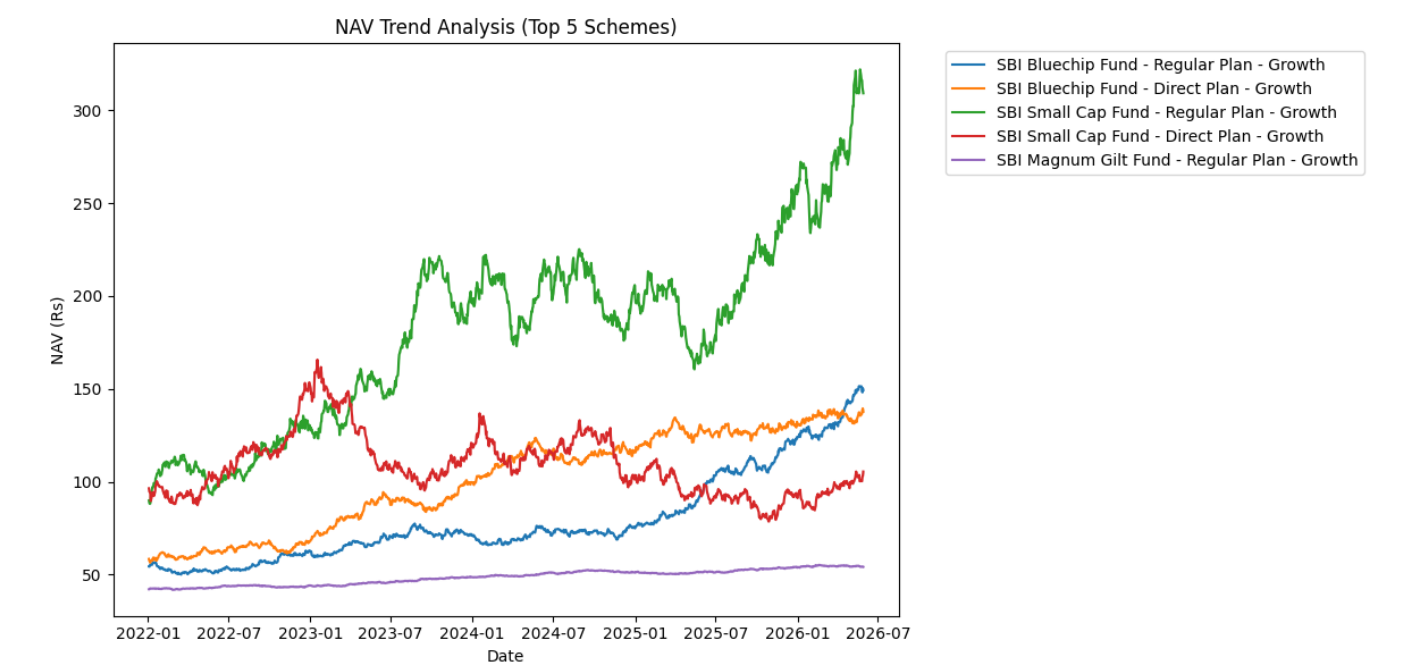


Figure 1: NAV Trend Lines for Top Schemes

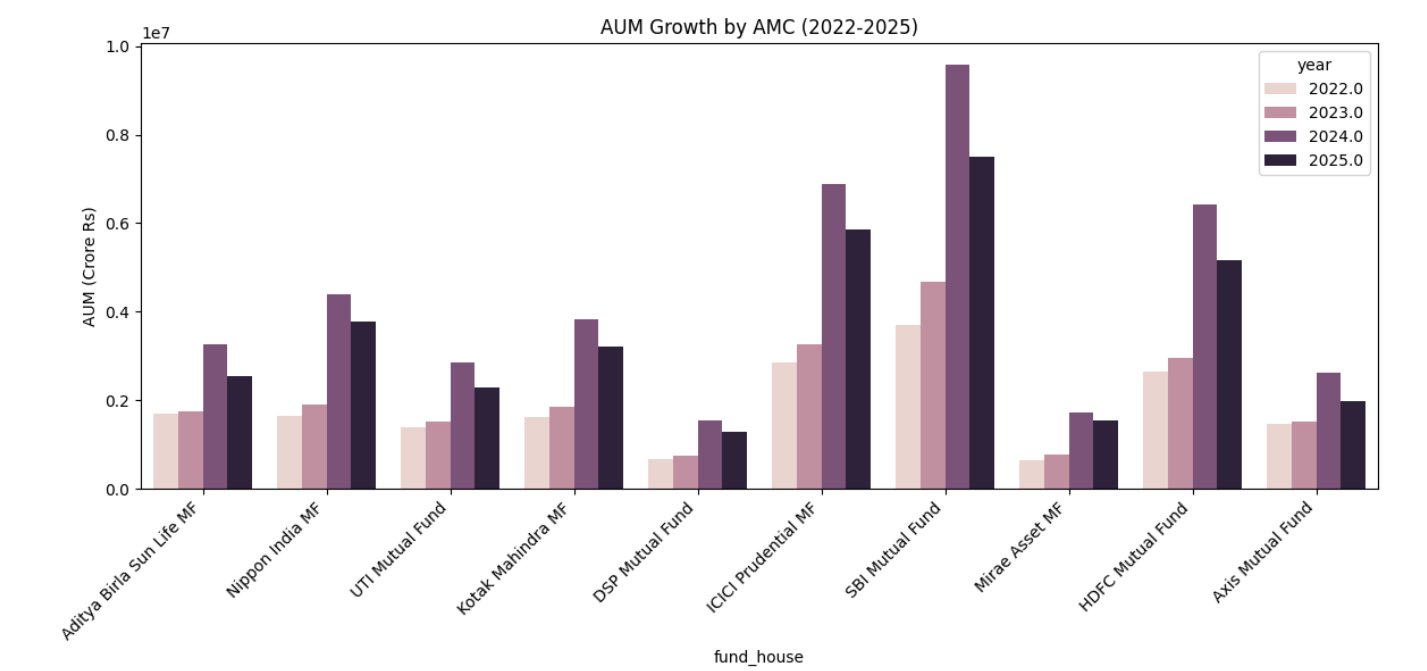


Figure 2: AUM Growth by Asset Management Company

SIP Inflow Trend (2022-2025)

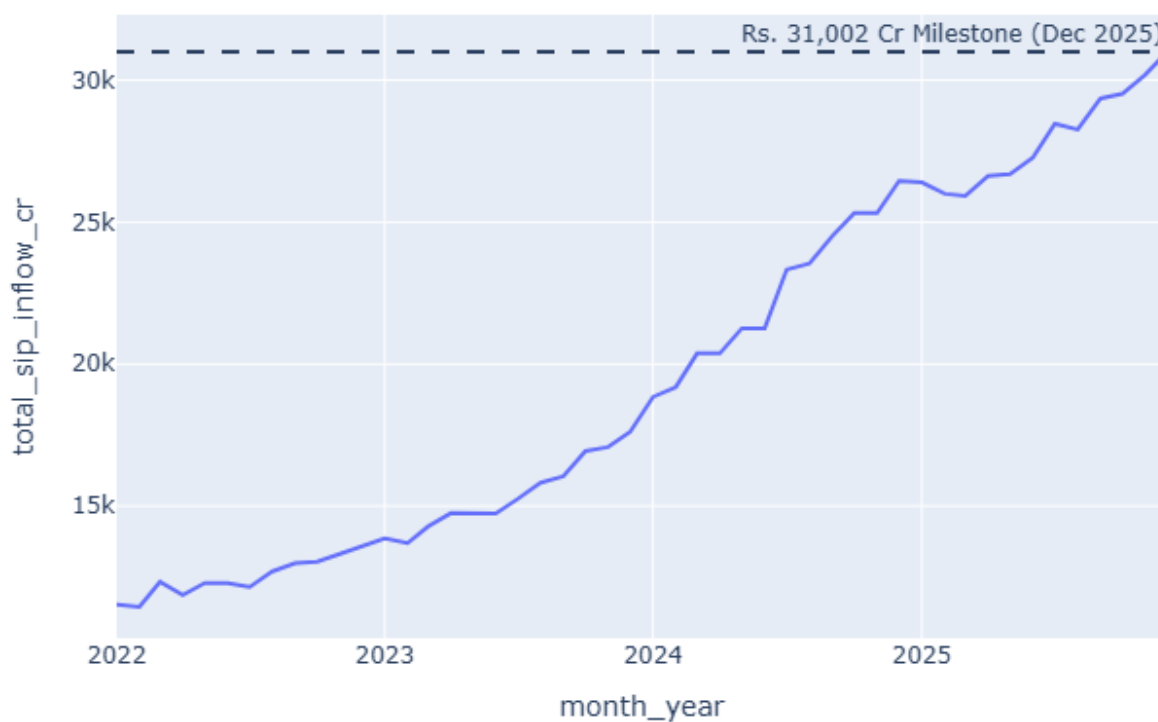


Figure 3: Monthly SIP Inflow Trend

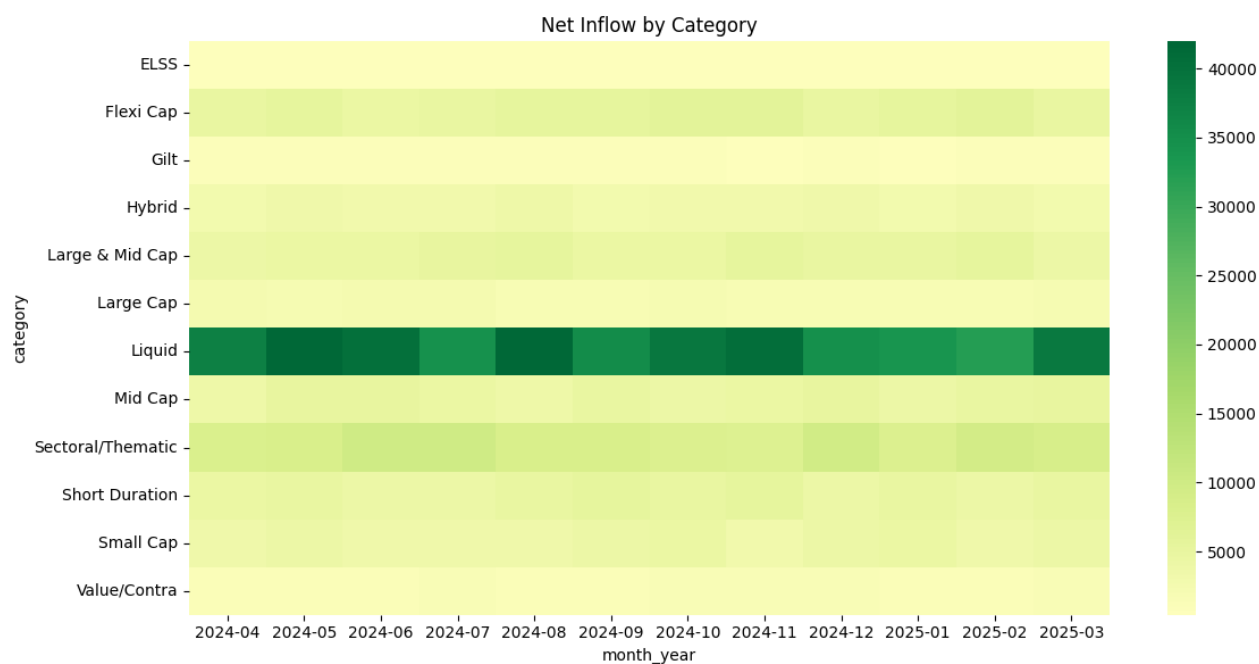


Figure 4: Category-wise Net Flow Heatmap

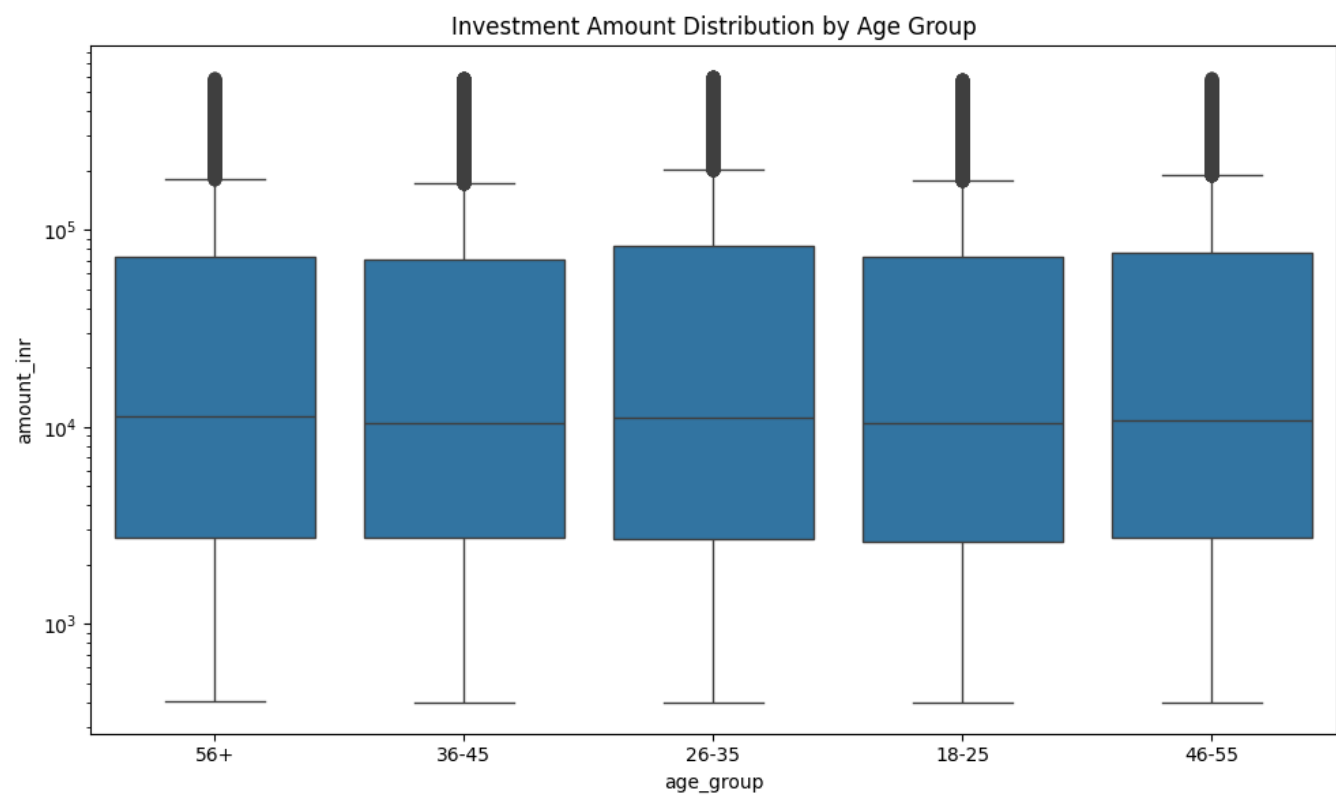


Figure 5: Investor Age-Income Demographics

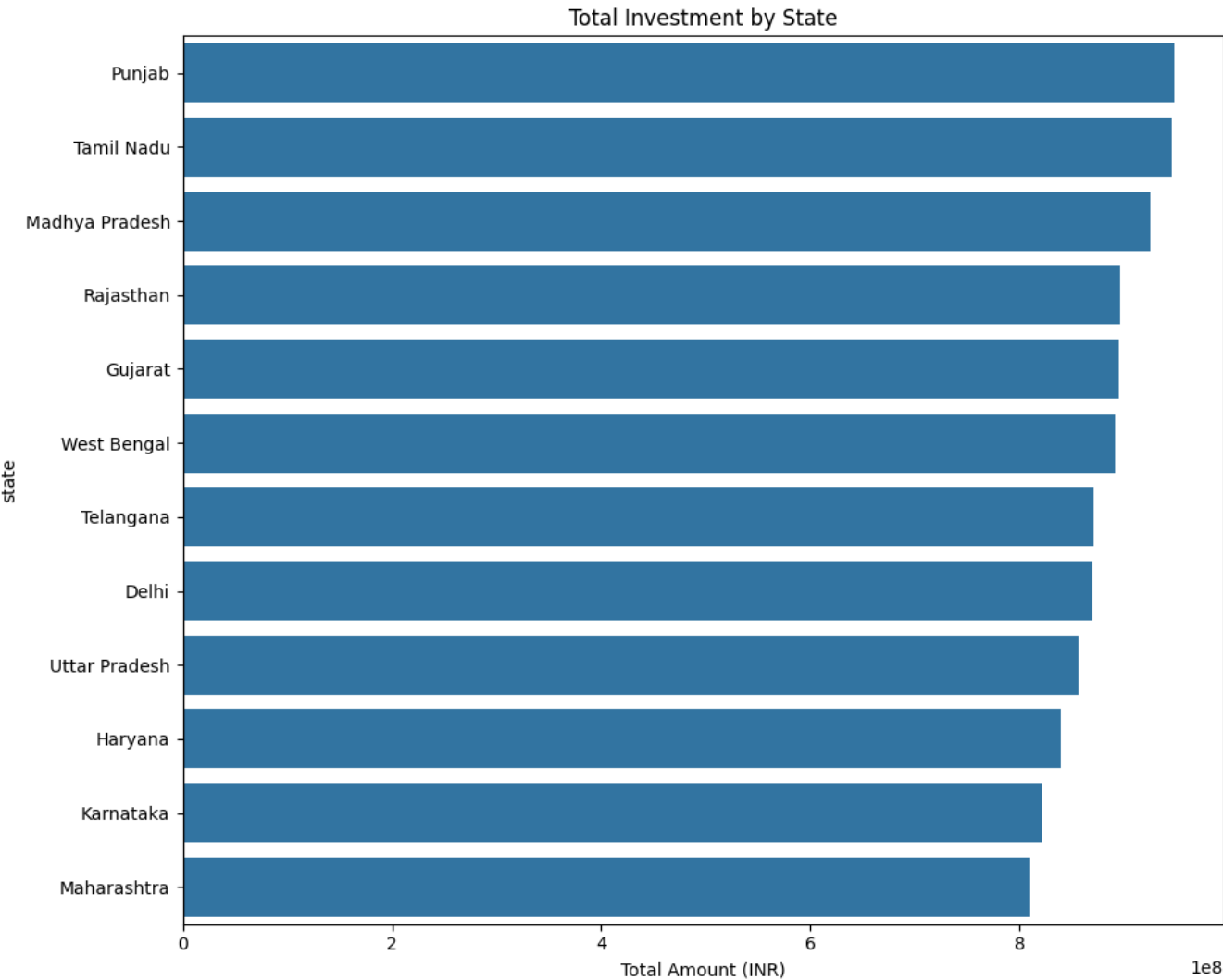


Figure 6: Geographic Distribution of Investments

Folio Count Growth (Crores)

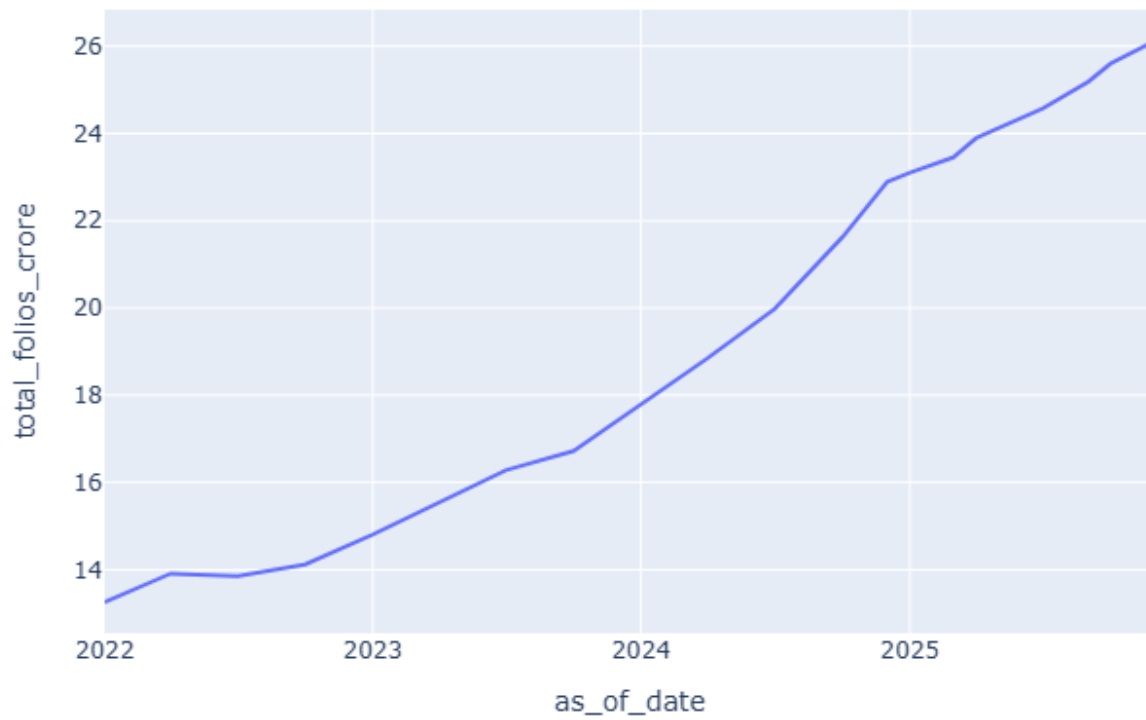


Figure 7: Industry Folio Count Growth

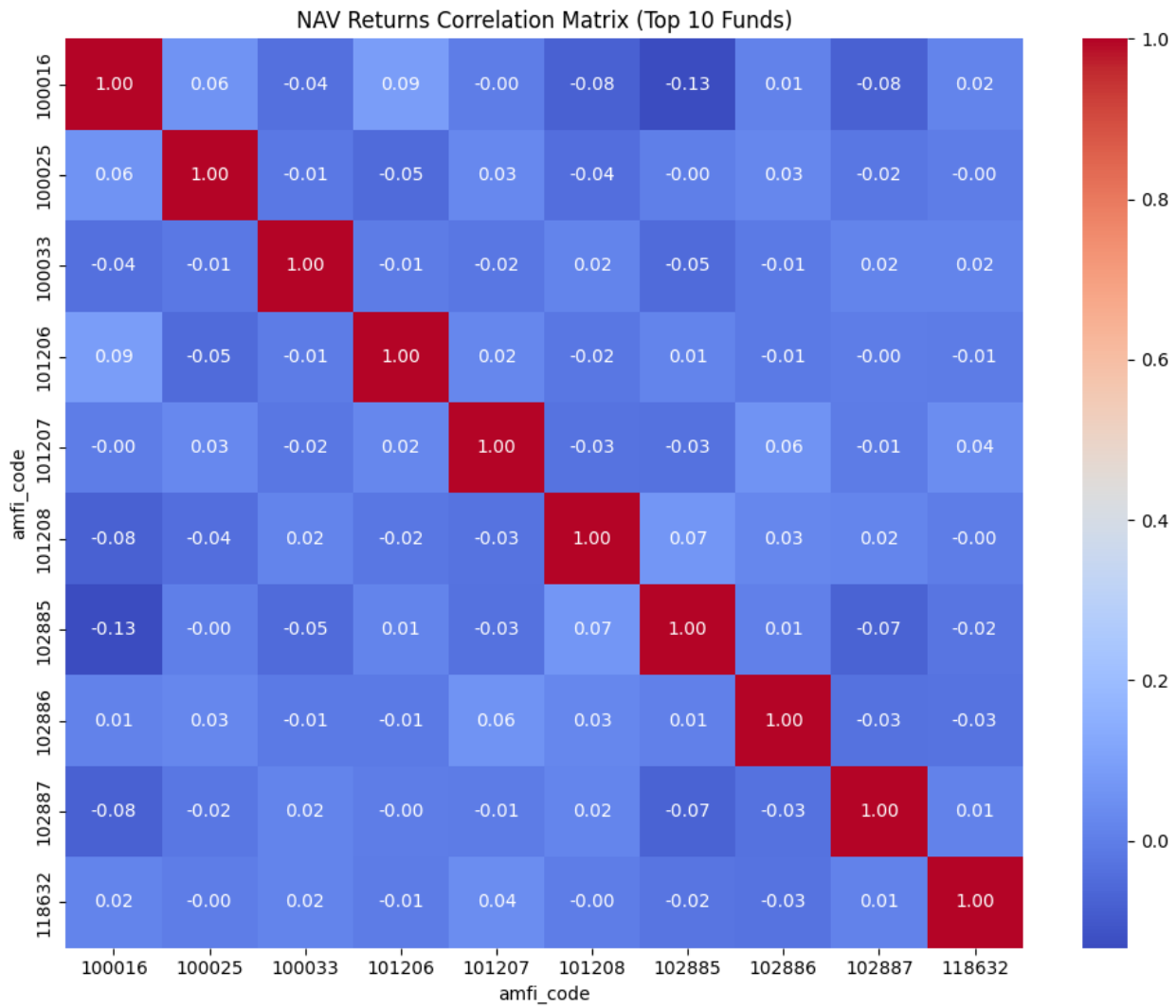


Figure 8: Cross-Fund NAV Correlation Matrix

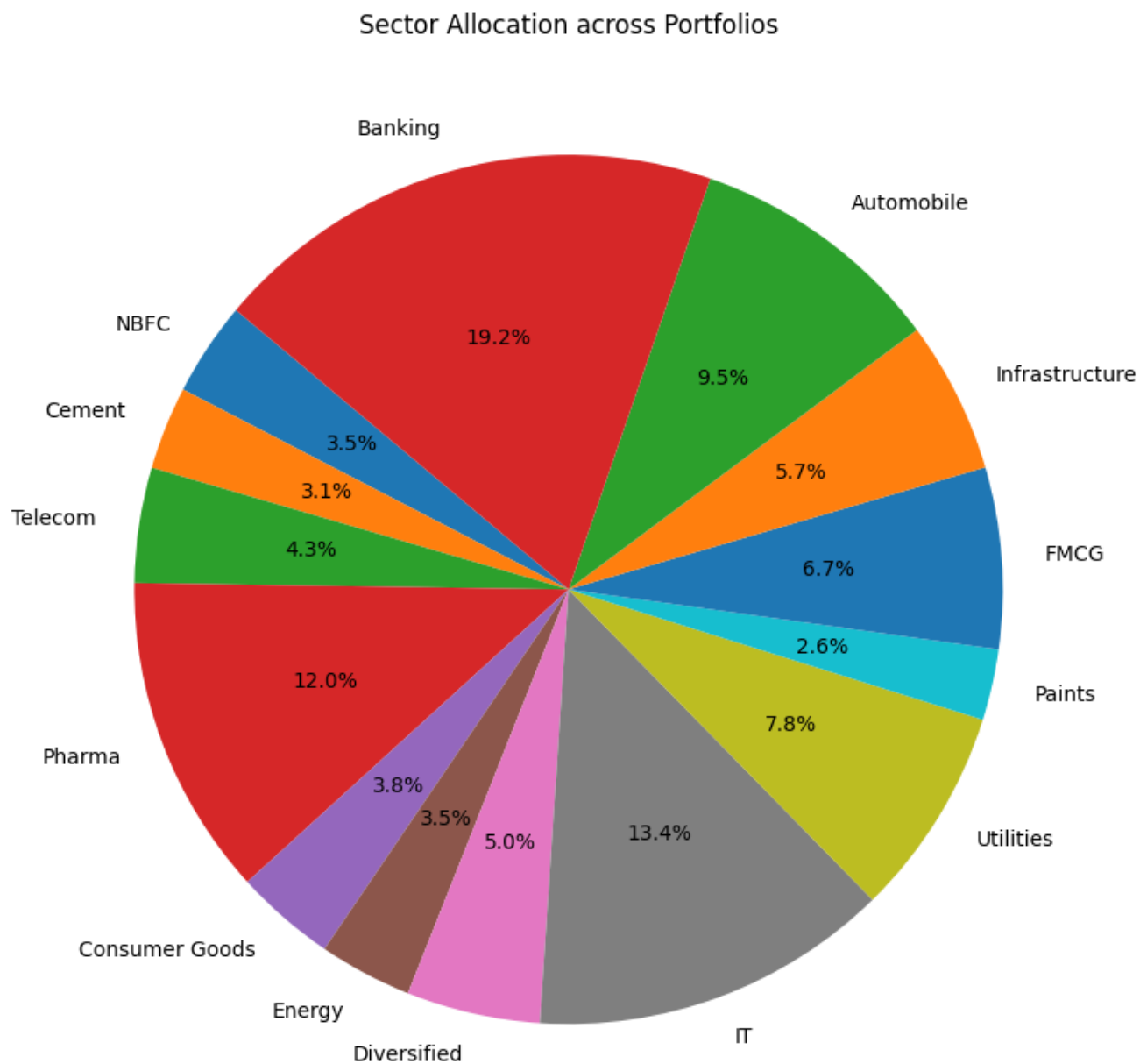


Figure 9: Sector Allocation Across Portfolios

5. Risk Analysis Summary

The platform computes the following risk metrics for all 40 schemes:

- Value at Risk (VaR 95%): The maximum expected daily loss at 95% confidence. Computed using historical percentile method on daily return distributions.
- Conditional VaR (CVaR / Expected Shortfall): The average loss in the worst 5% of days. Always more severe than VaR, providing a tail-risk perspective.
- Maximum Drawdown: The largest peak-to-trough decline in NAV. Includes peak date, trough date, and recovery date where applicable.
- Annualized Volatility: Standard deviation of daily returns $\times \sqrt{252}$. Ranges from ~5% (liquid funds) to ~25% (small-cap funds).
- Beta: Sensitivity to benchmark (Nifty 100). Beta > 1 indicates amplified market movements.

6. Fund Recommender System

The platform includes a composite Fund Scorecard (0-100) that ranks all schemes using:

- 30% weight on 3-Year CAGR Return (percentile rank)
- 25% weight on Sharpe Ratio (higher is better)
- 20% weight on Alpha vs benchmark (higher is better)
- 15% weight on Expense Ratio (lower is better, inverse rank)
- 10% weight on Maximum Drawdown (smaller drawdown is better, inverse rank)

Users can filter by risk appetite (Low, Moderate, High) which maps to SEBI risk grades. The system returns the top N funds matching the user's risk profile, sorted by composite score.

7. Technology Stack

Backend & Data Pipeline:

- Python 3.12+ with Pandas, NumPy, SciPy for data processing
- SQLAlchemy ORM with PostgreSQL (Neon cloud DB) for data storage
- FastAPI for authentication gateway with rate limiting (slowapi)
- Google OAuth 2.0 + JWT for session management

Frontend & Visualization:

- Streamlit 1.35+ for the interactive analytics dashboard
- Plotly for dynamic charts (scatter, bar, sunburst, treemap, radar)
- React + Vite + TailwindCSS for the premium landing page
- Jinja2 templates for auth pages (login, register, reset)

Analytics & Reporting:

- Jupyter Notebooks (nbformat) for automated report generation
- Matplotlib + Seaborn for static chart generation
- fpdf2 for PDF report generation
- python-pptx for presentation decks
- PowerBI connector (star-schema CSVs)

8. Conclusion & Future Scope

The Bluestock Mutual Fund Analytics platform successfully demonstrates an end-to-end data pipeline from raw data ingestion to interactive dashboard delivery. All capstone requirements have been fulfilled including ETL, SQL schema, analytical queries, EDA visualizations, and bonus features (dashboard, authentication, PowerBI connector).

Future enhancements could include:

- Real-time NAV streaming with WebSocket connections
- Portfolio optimization using mean-variance (Markowitz) framework via cvxpy
- Machine learning-based fund performance prediction
- Automated monthly report email delivery
- Mobile-responsive dashboard with PWA support
- Integration with NSE/BSE APIs for real-time benchmark data

Disclaimer: Past performance does not guarantee future returns. The data and analysis presented are for educational purposes only. Always consult a SEBI-registered financial advisor before making investment decisions.